

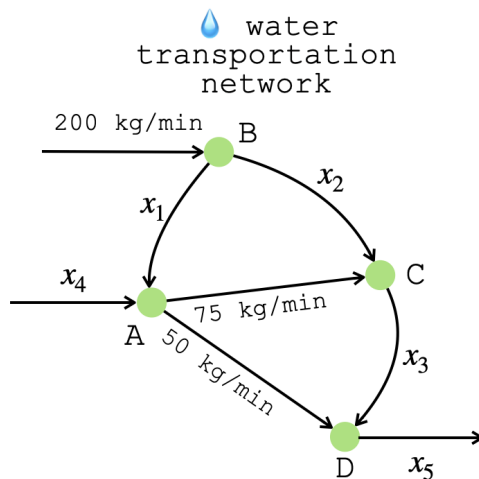
readings

1. this course concerns mathematical tools to solve problems in engineering. an important component of the course is learning to frame a problem and make appropriate abstractions and approximations to tackle the problem. with that in mind, please read the short essays “a systematic approach to modeling” by chemical engineer James Riggs [1] and “why model?” [2] by epidemiologist Joshua Epstein. the PDFs are posted on Canvas.
 - (a) provide a critique of the essay of James Riggs.
 - (b) provide a critique of the essay of Joshua Epstein.

word problems

for each problem, be sure to sketch a picture!

2. for making white wine, freshly picked grape clusters are de-stemmed continuously via a de-stemming machine. *de-stemming* grape clusters refers to the process of separating the berries from the stems. suppose 100 kg/hr of grape clusters are fed continuously into a grape de-stemming machine. write a system of two equations for the outlet flow rate of stems and of berries via (i) an overall mass balance and (ii) using the data that the grape clusters are 7% stems by weight [3]. write the equations in matrix form. find the solution via Gaussian elimination. to see the row picture, draw the two equations as lines (ideally, in Python or Julia) and ensure the intersection corresponds with your solution. (assume a perfect separation.)
solution: 7 kg/hr stems, 93 kg/hr berries.
3. a cooling crystallizer is used to form KNO_3 crystals from a solution of dissolved salt in water at 100°C . when 100 kg of solution is fed to the crystallizer, which operates at 30°C , 25 kg of solid KNO_3 is extracted. by how many degrees Celsius is the inlet stream superheated? use the solubility data on Wikipedia (interpolate.)
solution: see LearnChemE video.
4. a pipeline network for transporting water operates at steady state. there are four nodes that cannot accumulate water. write a system of linear equations, in matrix form, governing the five unknown flows in the network. find the general solution for these flows. what is the constraint on the flow out of x_5 to keep the flows in the directions shown?



solution: see James Hamblin's lecture 14.

5. write a system of linear equations in matrix form to systematically balance the chemical equation describing the combustion of ethanol:



use Gaussian elimination to solve the linear system.

solution: see Think Linear Algebra.

References

- [1] Riggs JB. A systematic approach to modeling. Chemical Engineering Education. 1988 22(1): 26-9.
- [2] Epstein JM. Why model? Journal of artificial societies and social simulation. 2008 11(4): 12.
- [3] Blackford M, Comby M, Zeng L, Dienes-Nagy A, Bourdin G, Lorenzini F, Bach B. A review on stems composition and their impact on wine quality. Molecules. 2021 26(5) :1240.